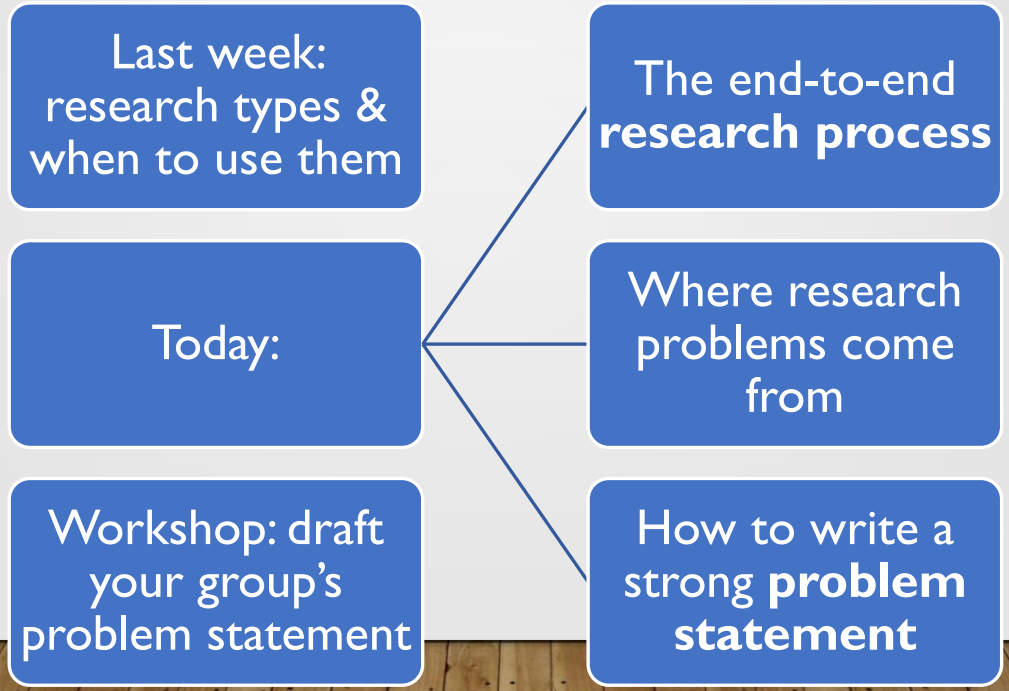
CSC4197

RESEARCH METHODOLOGY

THE RESEARCH PROCESS & PROBLEM IDENTIFICATION

2 RECAP & TODAY'S GOALS



3 THE RESEARCH PROCESS PIPELINE

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Idea → Problem → Literature Review → Gap → Research Question → Hypothesis → Research Design → Data Collection → Analysis → Interpretation → Writing → Dissemination → (new ideas...)
```

- **Iterative**, not strictly linear — the literature review often reshapes the problem
- Each stage has quality gates: a weak problem cannot be rescued by strong analysis

4 STAGE TIME BUDGET (REALITY CHECK)

Typical undergraduate thesis effort distribution:

- Problem + literature review: **25%**
- Design + data collection: **25%**
- Analysis: **20%**
- Writing + revisions: **25%**
- Presentation/defense prep: **5%**

⚠️ Most students under-budget the literature review and writing — leading to weak gaps and rushed papers.

5 WHERE DO RESEARCH PROBLEMS COME FROM?

1. **Limitations sections** of recent papers ("future work" goldmine)
2. **Industry pain points** — internships, hackathons, freelance clients
3. **Replication & extension** — does method X work on Bengali text? On Bangladeshi data?
4. **Emerging technology** — LLMs, edge AI, post-quantum crypto, AR/VR
5. **Societal needs / SDGs** — agriculture, health, education, traffic in BD context
6. **Your own frustration** — "why is there no good tool for ___?"

6 THEORY: THE FINER CRITERIA

A good research problem is:

- **F**easible — data, skills, equipment, time available (one semester! 6 months!)
- **I**nteresting — to you *and* the community
- **N**ovel — not already fully answered (verified via literature)
- **E**thical — no harm; approvals obtainable
- **R**elevant — advances knowledge or practice

Quick test: Can you explain in 2 sentences why anyone should care? If not, refine.

7 FINER IN ACTION — EVALUATING A PROPOSED PROBLEM

- **Proposed problem:**
- *"We'll build an app that uses AI to predict dengue outbreaks in Dhaka and alerts every resident before an outbreak happens."*

	Criterion	Verdict	Why
F	Feasible	Weak	"Every resident," real-time alerts → no access to citywide health data or delivery channel in one semester
I	Interesting	Pass	Dengue is a locally urgent, motivating problem
N	Novel	Weak	Novelty claimed, not shown against prior dengue-prediction work
E	Ethical	Weak	Health alerts → false alarms cause panic; needs consent & accuracy safeguards
R	Relevant	Pass	Directly serves a Bangladesh public-health need (SDG-aligned)

8 FINER IN ACTION — EVALUATING A PROPOSED PROBLEM

- **Tightened version (now passes):**
- *"This study evaluates whether weekly dengue case counts for Dhaka can be predicted 2 weeks ahead from public DGHS surveillance data and weather variables, comparing an LSTM model against an ARIMA baseline."*

- *FINER fails are usually fixed by narrowing scope (F), grounding novelty in a gap (N), and using public/aggregate data (E), not by abandoning the topic.*

9 NARROWING THE FUNNEL: TOPIC → PROBLEM

Level	Example
Broad area	Machine learning in healthcare
Topic	ML for ICU outcome prediction
Problem	Mortality prediction models are accurate but poorly **calibrated**, so clinicians distrust the probabilities
Researchable problem	"Existing ICU mortality models trained with oversampling produce miscalibrated risk estimates on imbalanced real-world data"

Each step adds **specificity**: population, context, deficiency, consequence.

10 ANATOMY OF A PROBLEM STATEMENT

A strong problem statement has 4 moves (one paragraph):

1. **Context** — establish the domain and why it matters
2. **Ideal** — what should be true / what is desired
3. **Reality + deficiency** — what is actually true; what's missing or broken
4. **Consequence** — what happens if unsolved; who is affected

Template: *"In [domain], [ideal]. However, [current limitation/gap]. As a result, [negative consequence]. This research addresses [specific problem]."*

11 WORKED EXAMPLE (CS PROBLEM STATEMENT)

"Online code submission platforms are central to programming education in Bangladeshi universities **(context)**. Ideally, automated feedback should help students locate logical errors quickly **(ideal)**. However, current autograders report only pass/fail verdicts without explanation, and existing LLM-based feedback tools have not been evaluated for first-year students writing in mixed Bangla-English contexts **(deficiency)**. Consequently, struggling students disengage and instructor workload increases **(consequence)**. This research designs and evaluates an LLM-assisted feedback layer for introductory programming courses."

✓ **Identify the 4 moves — and the implied research type.**

12 COMMON PITFALLS IN PROBLEM IDENTIFICATION

- **Solution in search of a problem** ("I want to use blockchain" — for what deficiency?)
- **Too broad** ("improve cybersecurity in Bangladesh")
- **Already solved** — discovered too late because lit review was skipped
- **Unverifiable claims** ("no one has ever studied X" — dangerous; say "to the best of our knowledge")
- **Infeasible scope** (national-scale data collection in one semester)

13 EXERCISE



https://tinyurl.com/finer-exercise

14 HANDS-ON WORKSHOP (~60 MIN)

Step 1 (15 min): Each group brainstorms 3 candidate problems using the 6 sources from Slide 5

Step 2 (15 min): Apply FINER — score each candidate 1–5 per criterion; keep the winner

Step 3 (20 min): Write a 4-move problem statement

Step 4 (10 min): Swap with a neighboring group → peer critique: *Can you find all 4 moves? Is it FINER?*

Deliverable: problem statement will be submitted as hardcopy in next class.

15 WRAP-UP & NEXT WEEK

Key takeaways - Research is a pipeline; the problem stage determines everything downstream - Use FINER to filter; use the 4-move template to articulate - Specificity = population + context + deficiency + consequence

Next week: Literature review & research-gap identification — bring laptops with university library/IEEE Xplore access.

Homework: Read Keshav's three-pass method; find 3 papers related to your group's problem